Spring 1989

Problem 1 (Sp89). Let $a_1, a_2, \ldots$ be positive numbers such that $\sum_{n=1}^{\infty} a_n < \infty$. Prove that there are positive numbers $c_1, c_2, \ldots$ such that

$$\lim_{n\to\infty} c_n = \infty \quad \text{and} \quad \sum_{n=1}^{\infty} c_n a_n < \infty$$

Problem 2 (Sp89). Let $\mathbb{F}$ be a field, n and m positive integers, and A an $n \times n$ matrix with entries in $\mathbb{F}$ such that $A^m = 0$. Prove that $A^n = 0$.

Problem 3 (Sp89, Fa06, Sp23). Let $f\colon \mathbb{R} \times [0,1] \to \mathbb{R}$ be a continuous function. For $x \in \mathbb{R}$, define

$$g(x) = \max\{f(x,y) \mid y \in [0,1]\}$$

Show that g is continuous.

Problem 4 (Sp79, Su85, Sp89, Fa19). Prove that if $1 < \lambda < \infty$, the function

$$f_\lambda(z) = z + \lambda - e^z$$

has only one zero in the half-plane $\Re z < 0$, and that this zero is real.

Problem 5 (Sp89, Fa24). Let G be a group whose order is twice an odd number. For g in G, let λ_g denote the permutation of G given by $\lambda_g(x) = gx$ for $x \in G$.

1. Let g be in G. Prove that the permutation λ_g is even if and only if the order of g is odd.
2. Let $N = \{g \in G \mid \operatorname{ord}(g) \text{ is odd}\}$. Prove that N is a normal subgroup of G of index 2.

Problem 6 (Sp89, Fa97, Fa04). Let S be the subspace of $M_{n\times n}$ (the vector space of all real $n\times n$ matrices) generated by all matrices of the form $AB - BA$ with A and B in $M_{n\times n}$. Prove that $\dim S = n^2 - 1$.

Problem 7 (Sp89). Let

$$A = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \qquad B = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Find the general solution of the matrix differential equation $\frac{\mathrm{d}X}{\mathrm{d}t} = AXB$ for the unknown 4×4 matrix function $X(t)$.

Problem 8 (Fa81, Sp89, Fa97). Let f be a holomorphic map of the unit disc $\mathbb{D} = \{z \mid |z| < 1\}$ into itself, which is not the identity map $f(z) = z$. Show that f can have, at most, one fixed point.

Problem 9 (Sp89). For G a group and H a subgroup, let $C(G, H)$ denote the collection of left cosets of H in G. Prove that if H and K are two subgroups of G of infinite index, then G is not a finite union of cosets from $C(G, H) \cup C(G, K)$.

Problem 10 (Sp89). Let the real $2n \times 2n$ matrix X have the form

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

where A, B, C, and D are $n \times n$ matrices that commute with one another. Prove that X is invertible if and only if $AD - BC$ is invertible.

Problem 11 (Sp89). Let R be a ring with at least two elements. Suppose that for each nonzero a in R there is a unique b in R (depending on a) with $aba = a$. Show that R is a division ring.

Problem 12 (Sp89). Evaluate $\displaystyle\int_C (2z - 1)e^{z/(z-1)}\, dz$ on the circle $|z| = 2$ with counterclockwise orientation.

Problem 13 (Sp89). Let D_{2n} be the dihedral group with $n \geqslant 3$. Determine its center $Z = \{c \in D_{2n} \mid cx = xc \text{ for all } x \in D_{2n}\}$.

Problem 14 (Sp89). Suppose f is a continuously differentiable map of $\mathbb{R}^2$ into $\mathbb{R}^2$. Assume that f has only finitely many singular points, and that for each positive number M, the set $\{z \in \mathbb{R}^2 \mid |f(z)| \leqslant M\}$ is bounded. Prove that f maps $\mathbb{R}^2$ onto $\mathbb{R}^2$.

Problem 15 (Sp89). Let $B = (b_{ij})_{i,j=1}^{20}$ be a real 20×20 matrix such that

$$b_{ii} = 0 \qquad \text{for } 1 \leqslant i \leqslant 20$$

$$b_{ij} \in \{1, -1\} \qquad \text{for } 1 \leqslant i, j \leqslant 20 \;\; i \neq j$$

Prove that B is nonsingular.

Problem 16 (Sp89, Sp00). Let f and g be entire functions such that

$$\lim_{z \to \infty} f(g(z)) = \infty$$

Prove that f and g are polynomials.

Problem 17 (Sp89). 1. Let R be a commutative ring with identity containing an element a with $a^3 = a + 1$. Further, let $\mathfrak{I}$ be an ideal of R of index < 5 in R. Prove that $\mathfrak{I} = R$.

2. Show that there exists a commutative ring with identity that has an element a with $a^3 = a + 1$ and that contains an ideal of index 5.

Problem 18 (Sp89). Let f be a real valued function on $\mathbb{R}^2$ with the following properties:

1. $f(K)$ is compact whenever K is a compact subset of $\mathbb{R}^2$.
2. For each y_0 in $\mathbb{R}$, the function $x \mapsto f(x, y_0)$ is continuous.
3. For each x_0 in $\mathbb{R}$, the function $y \mapsto f(x_0, y)$ is continuous.

Prove that f is continuous.
Note: See also **??**.